BDSD MPC Training Report

IDBS Reference: EXXXXX

Report Generated for TSLP_PF

Model Training Dataset: MR24-021

States in Model: IGG, VCC, Viability, Glutamate, Ammonium

Inputs in Model: Normalized_Bolus_Feed, Temperature_setpoint, pH_setpoint

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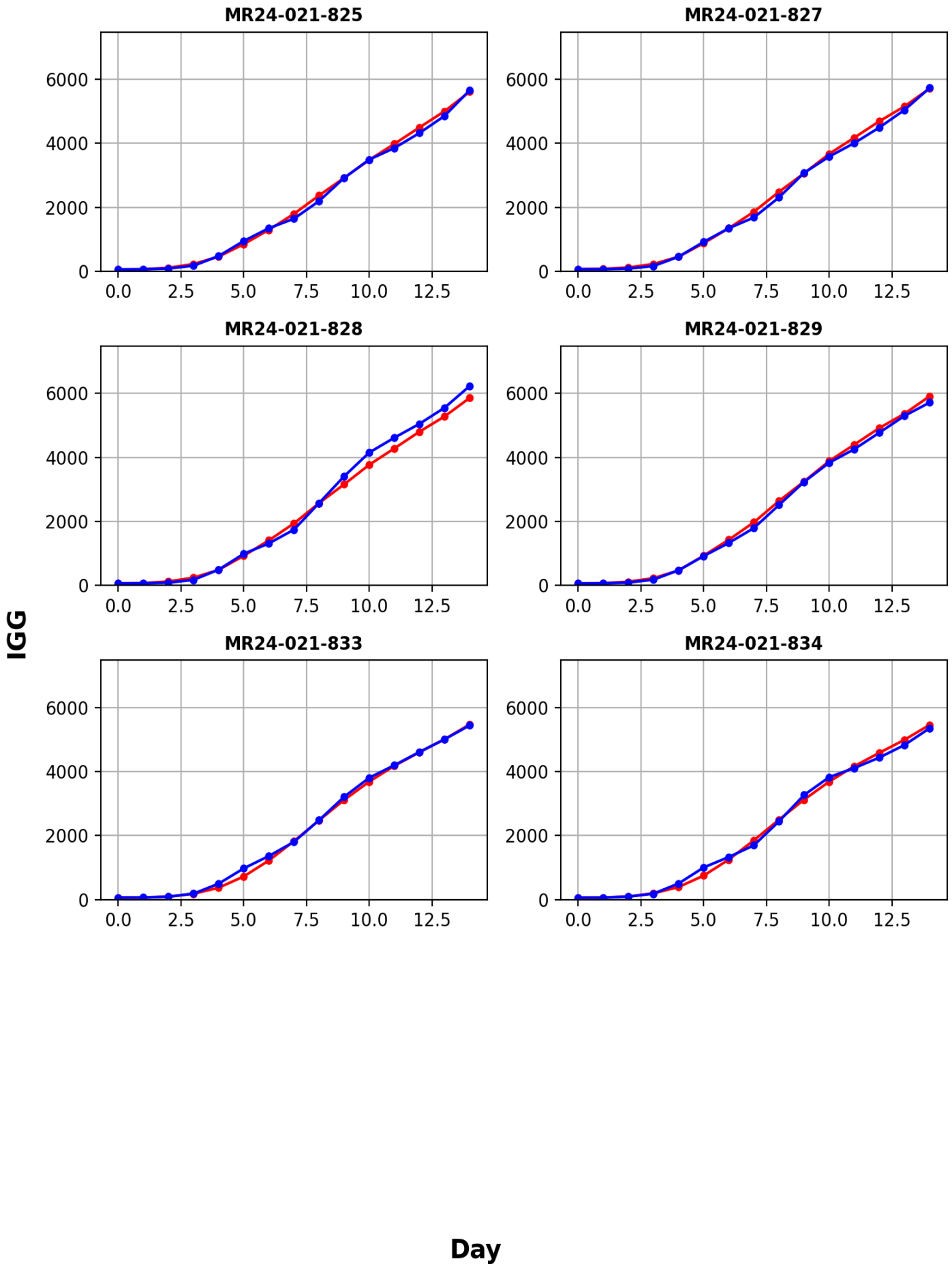
Report Date: 2025/10/06 17:12:18

Scaling Parameters

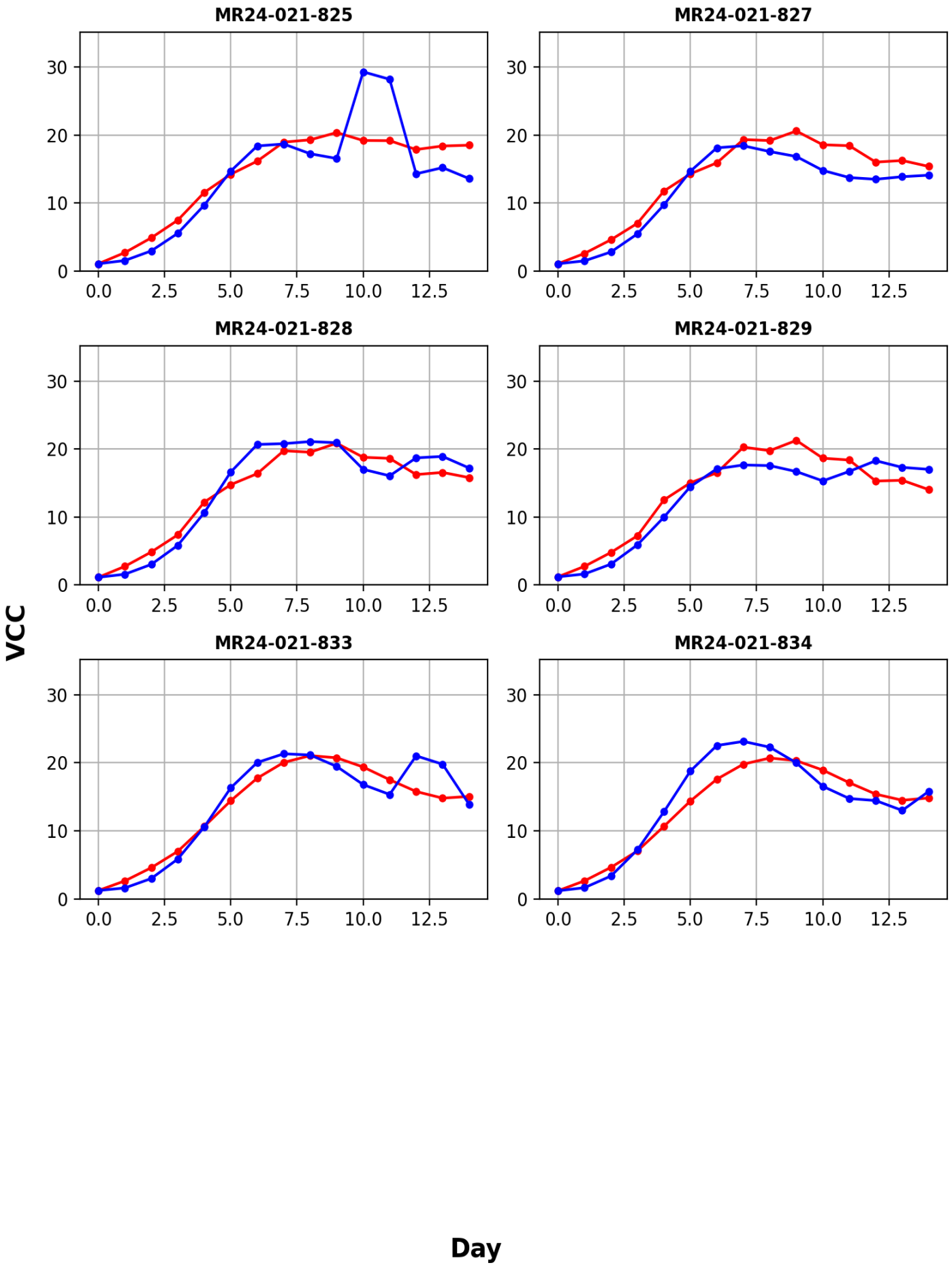
feature_names_in_	scale_	min_	data_min_	data_max_	data_range_
IGG	0.0	-0.01	58.674	6227.895	6169.221
VCC	0.035	-0.038	1.078	29.3	28.222
Viability	0.038	-2.831	73.6	99.6	26.0
Glutamate	0.101	-0.266	2.63	12.53	9.9
Ammonium	0.069	-0.037	0.54	14.98	14.44
Normalized_Bolus_Feed	14.433	0.0	0.0	0.069	0.069
Temperature_setpoint	1.0	-36.5	36.5	36.5	0.0
pH_setpoint	1.0	-7.0	7.0	7.0	0.0

Model Training Dataset

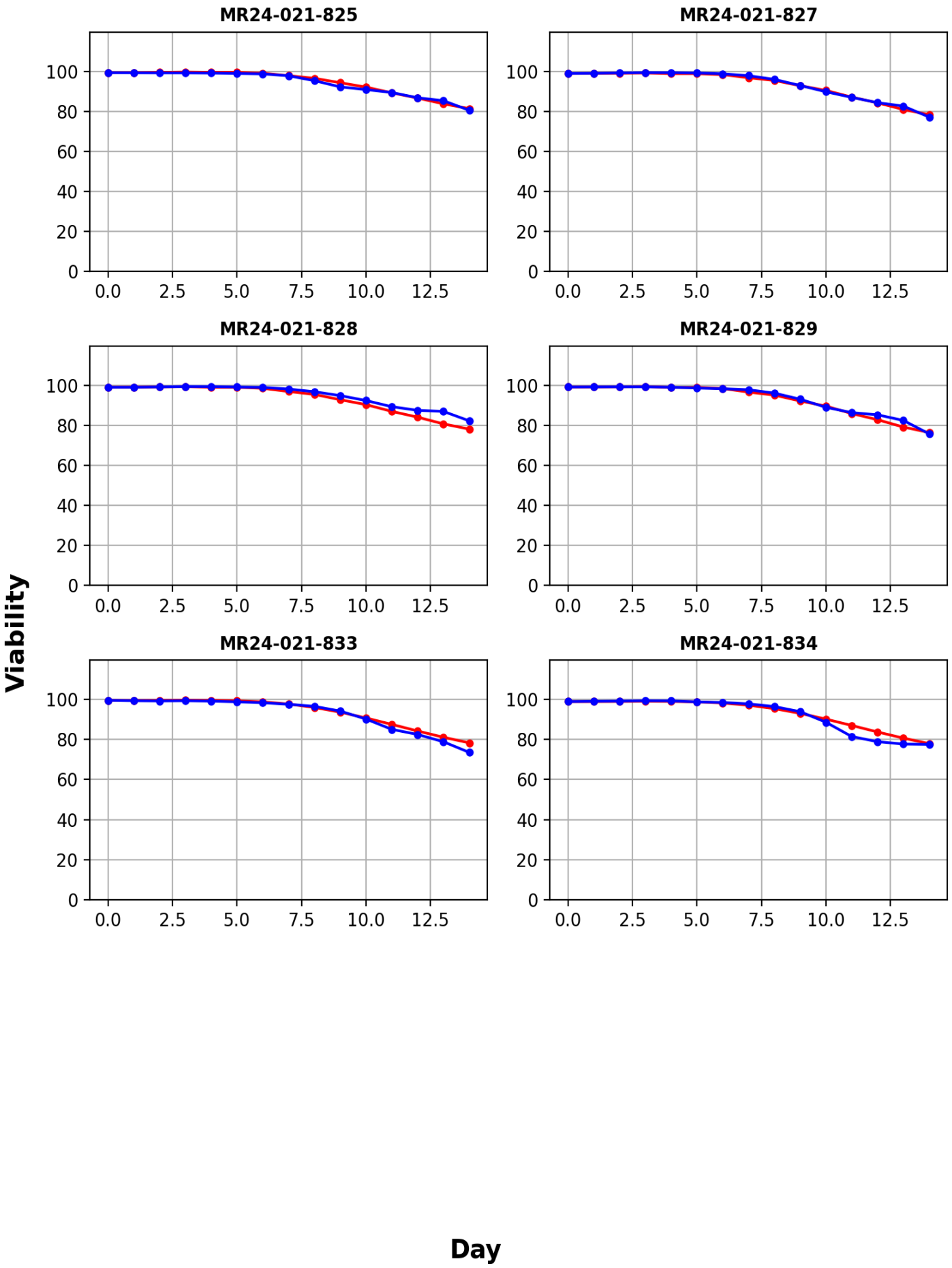
State: IGG


MR24-021-825
MR24-021-827
MR24-021-828
MR24-021-829
MR24-021-833
MR24-021-834
IGG
Day
0
2000
4000
6000
0.0
2.5
5.0
7.5
10.0
12.5

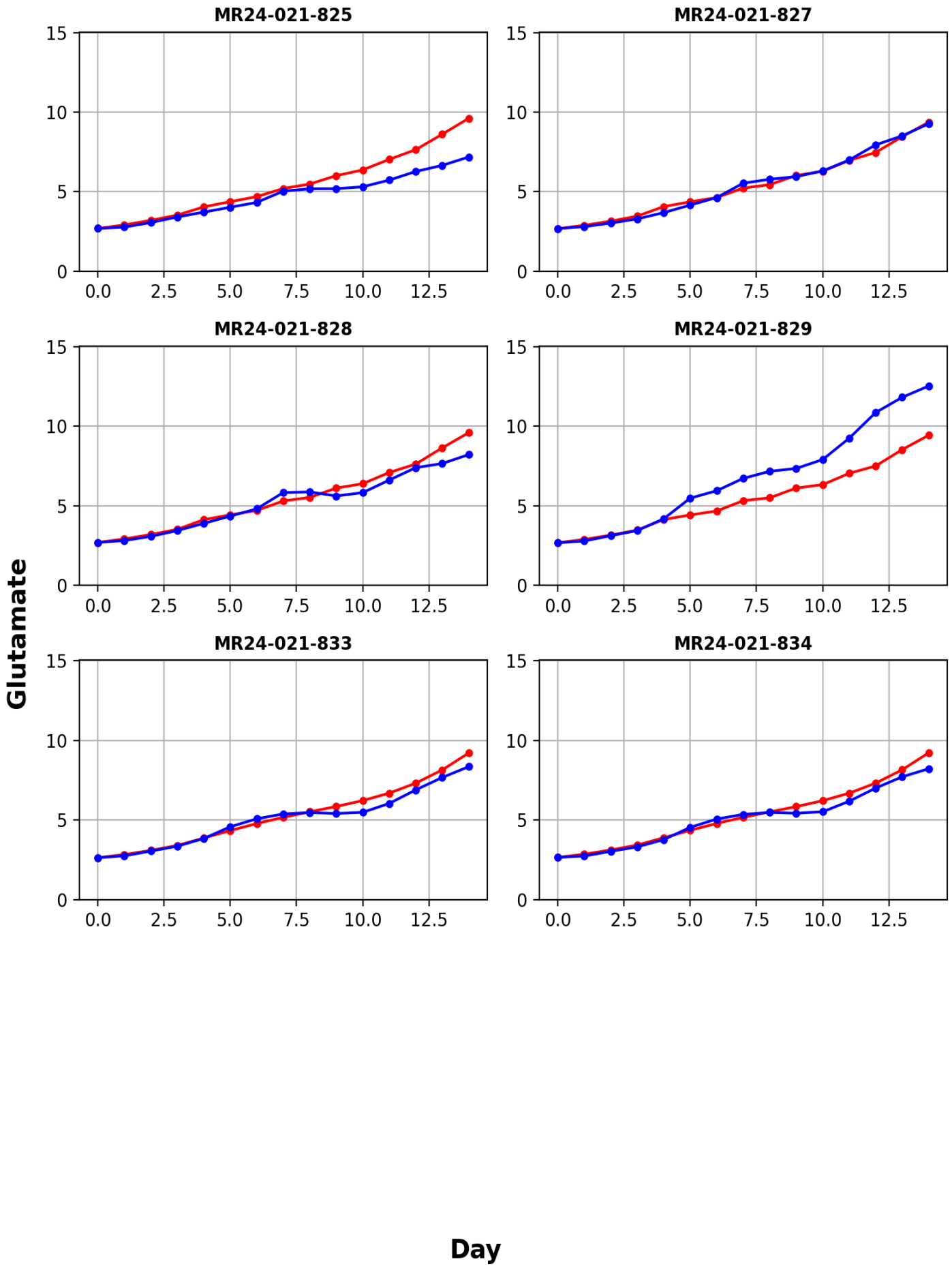
State: VCC


MR24-021-825
MR24-021-827
MR24-021-828
MR24-021-829
MR24-021-833
MR24-021-834
VCC
Day
0
10
20
30
0.0
2.5
5.0
7.5
10.0
12.5

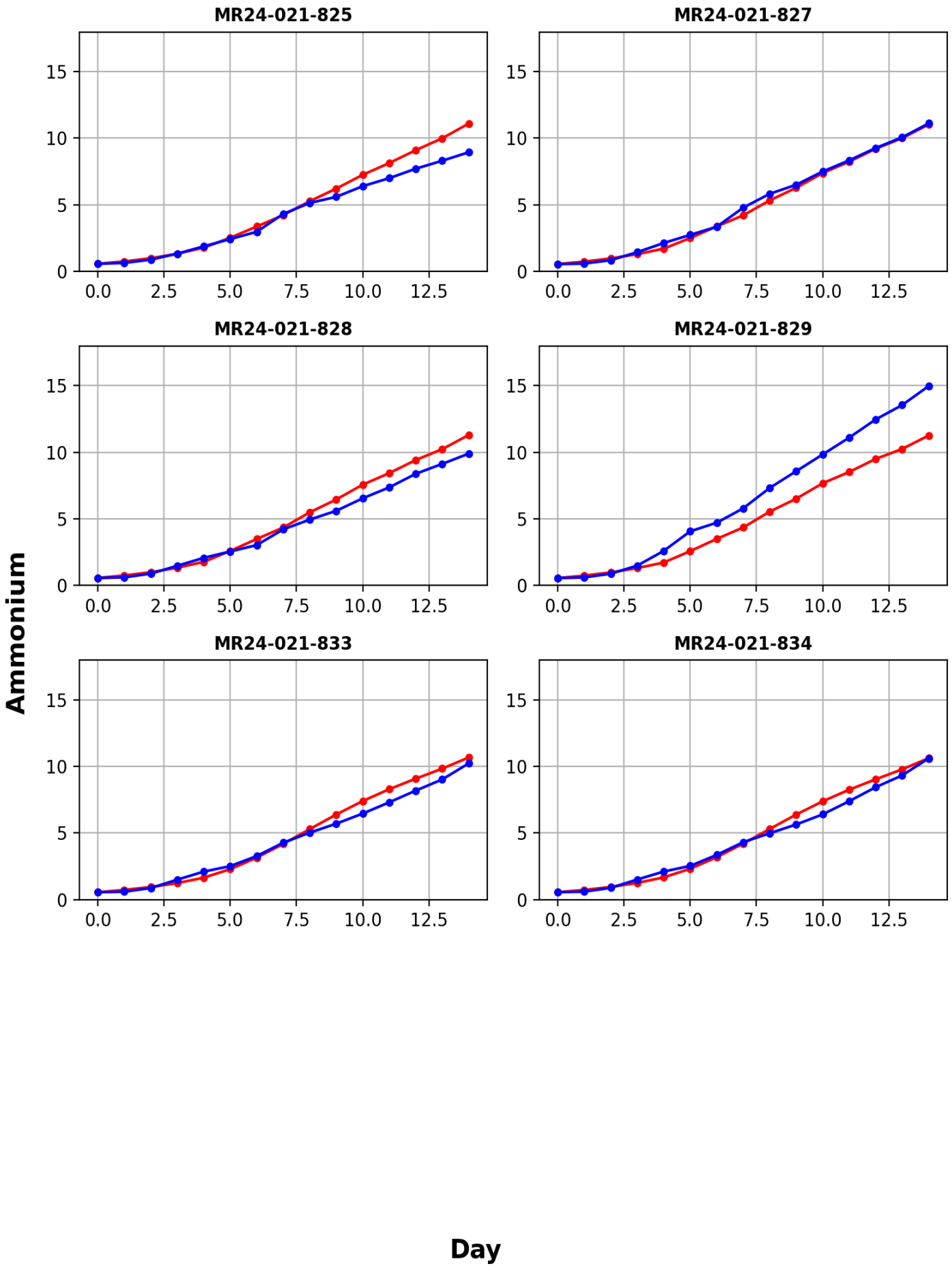
State: Viability



State: Glutamate

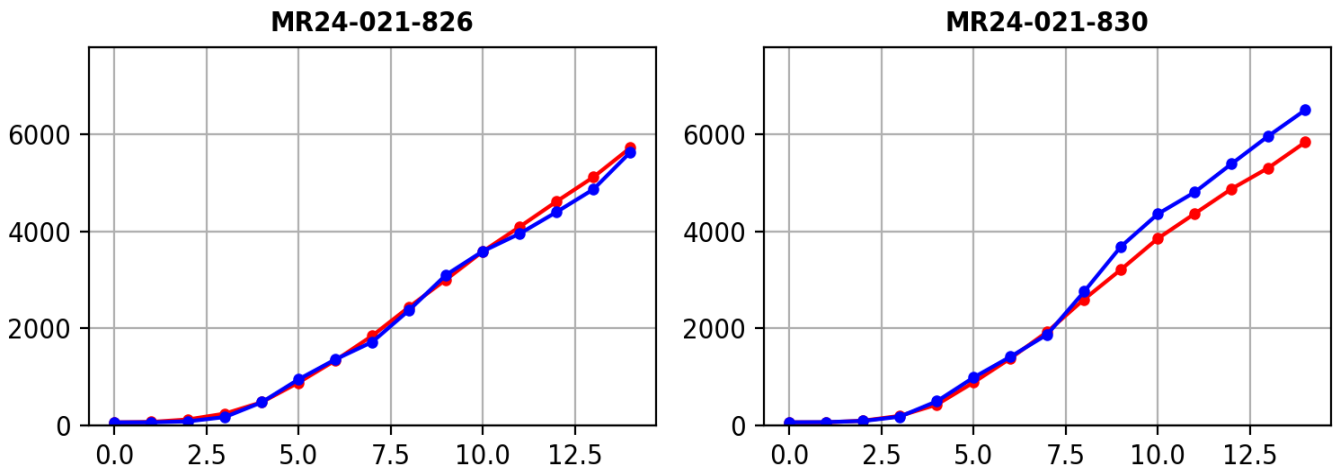


State: Ammonium



Model Testing Dataset

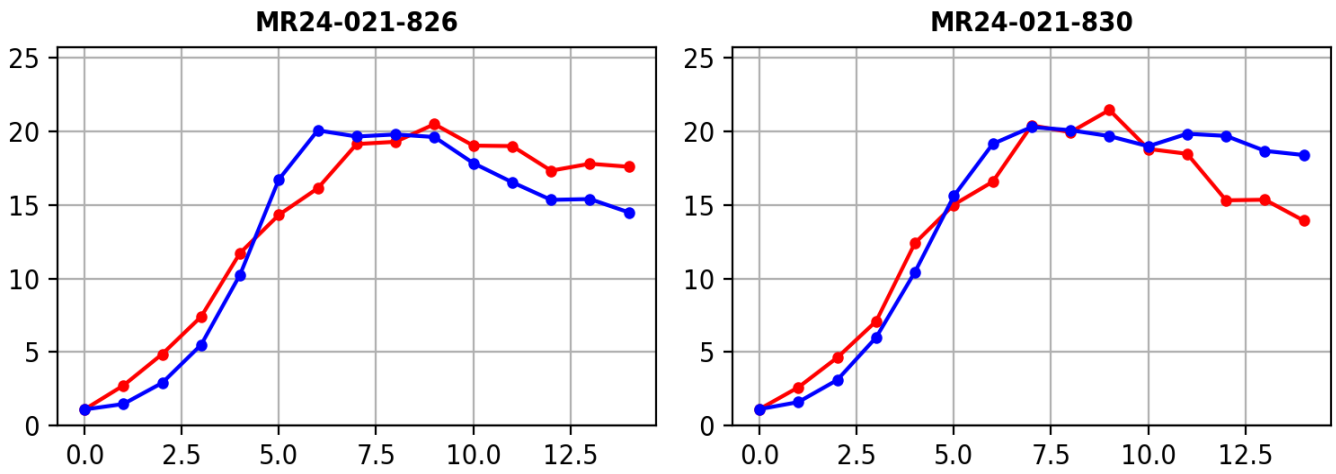
State: IGG



IGG

Day

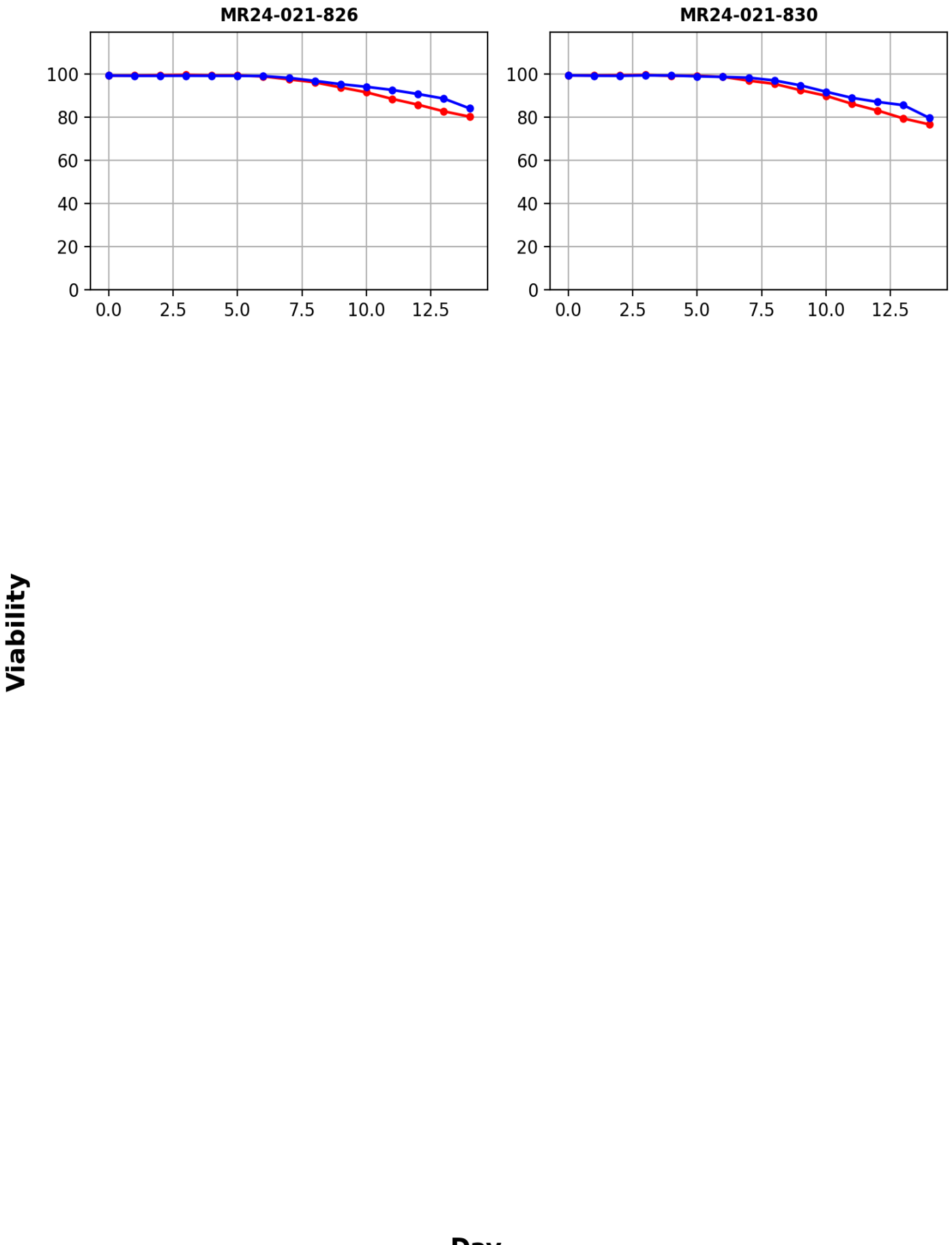
State: VCC



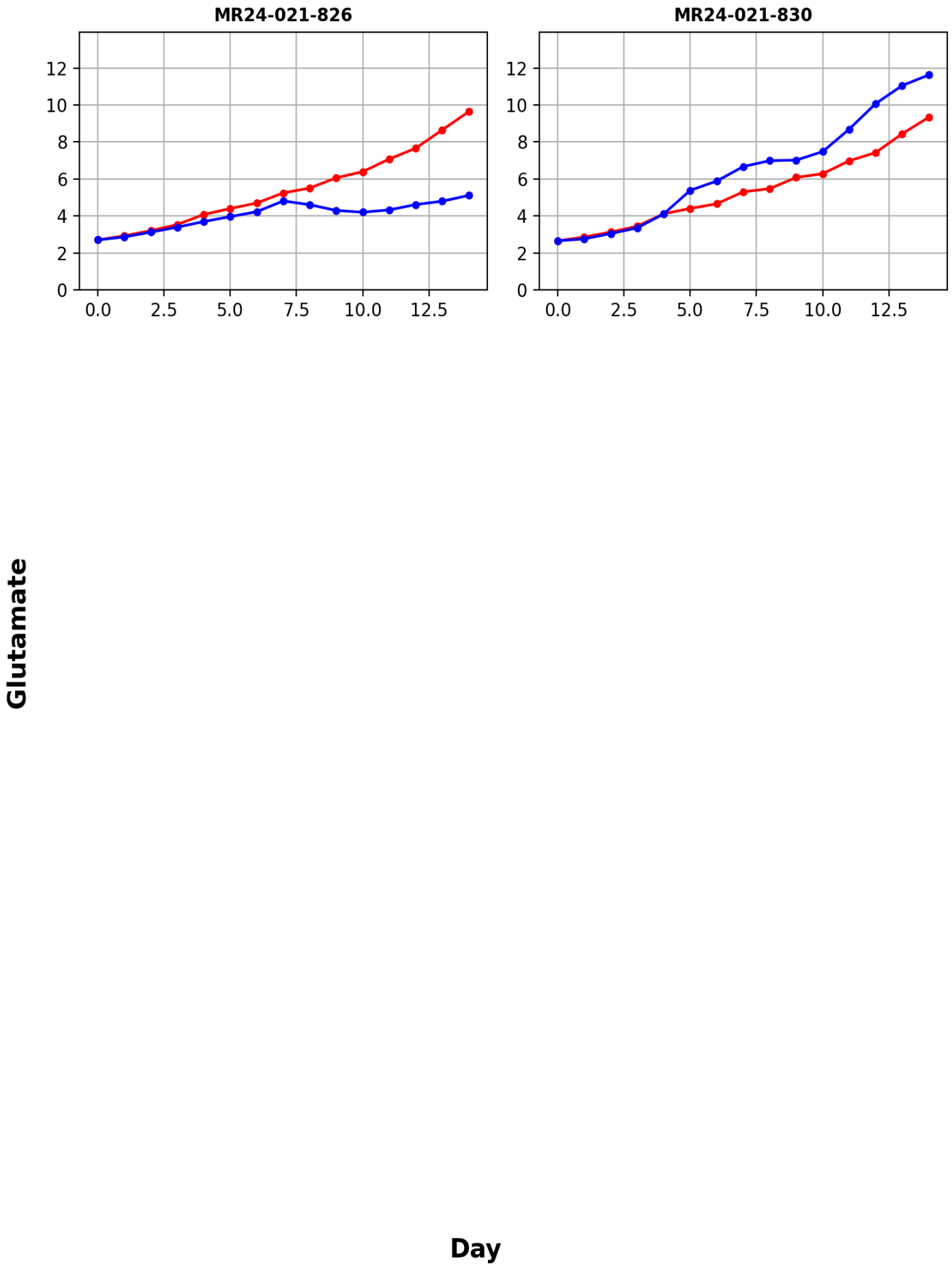
VCC

Day

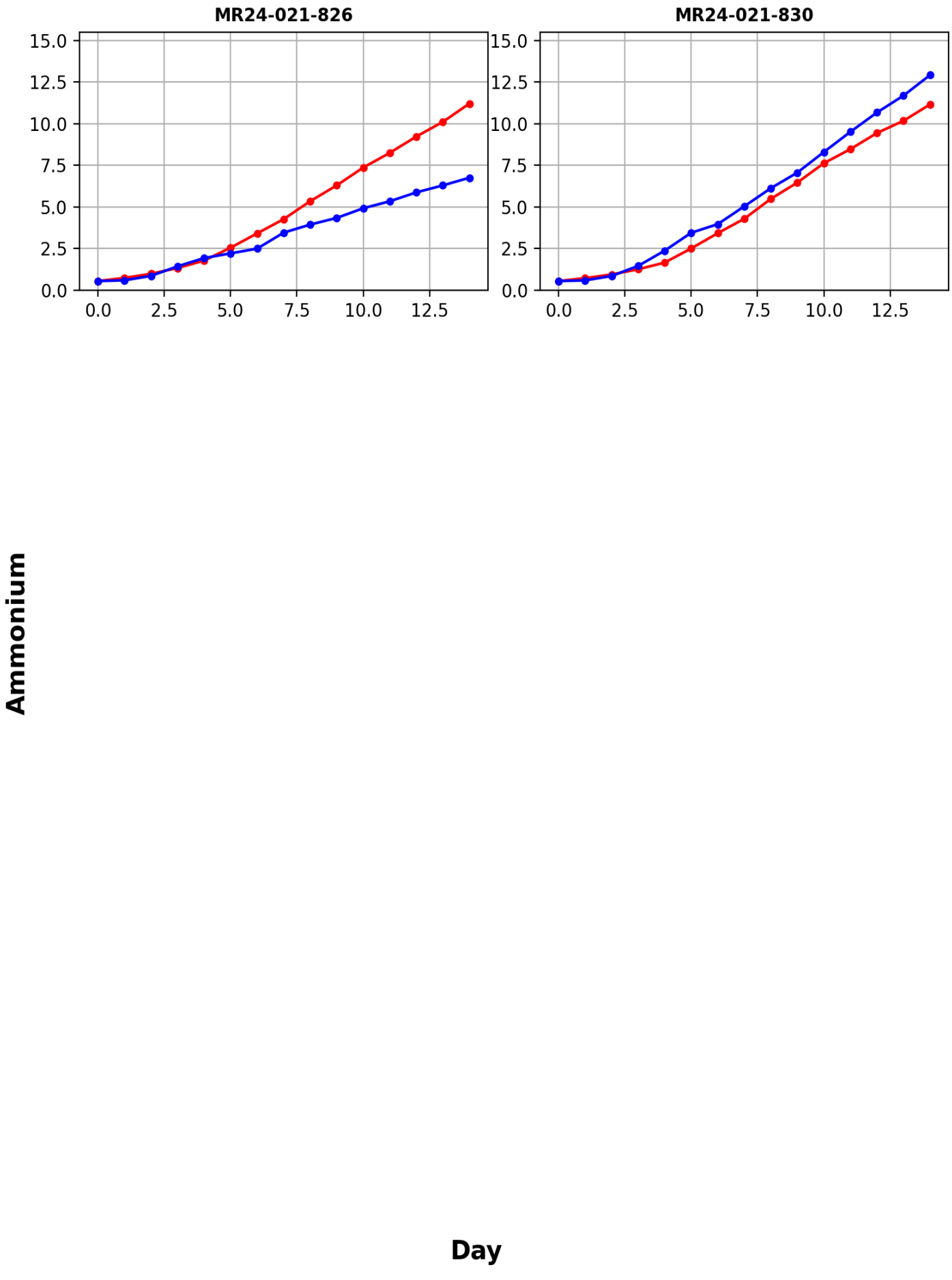
State: Viability



State: Glutamate



State: Ammonium



A Matrix

index	IGG	VCC	Viability	Glutamate	Ammonium
IGG	0.56741	0.18052	-0.00158	0.24329	-0.94237
VCC	-1.0	-0.12987	0.04217	1.0	0.38363
Viability	-0.13603	-0.03613	-0.00178	0.40128	-0.31181
Glutamate	-0.43243	-0.10155	0.01702	0.29893	0.47605
Ammonium	0.40631	0.11224	0.01168	0.15653	-0.64077

B Matrix

index	Normalized_Bolus_Feed	Temperature_setpoint	pH_setpoint
IGG	-0.0099	0.01565	-0.03075
VCC	0.10988	0.05651	-0.01271
Viability	-0.03347	0.03747	-0.10714
Glutamate	0.04433	-0.09074	-0.0537
Ammonium	-0.01587	0.04924	0.00694

Model RMSE Table

index	IGG	VCC	Viability	Glutamate	Ammonium
MR24-021-825	96.85	4.22	0.87	1.02	0.9
MR24-021-826	109.75	2.01	2.6	2.03	2.12
MR24-021-827	102.16	2.36	0.67	0.21	0.25
MR24-021-828	207.95	1.97	2.38	0.53	0.72
MR24-021-829	99.74	2.33	1.19	1.79	1.98
MR24-021-830	352.54	2.17	2.37	1.44	0.88
MR24-021-833	91.28	2.34	1.6	0.41	0.54
MR24-021-834	119.42	2.31	2.13	0.4	0.47

Model R2 Table

index	IGG	VCC	Viability	Glutamate	Ammonium
MR24-021-825	1.0	0.74	0.98	0.44	0.9
MR24-021-826	1.0	0.91	0.68	-6.98	-0.06
MR24-021-827	1.0	0.84	0.99	0.99	0.99
MR24-021-828	0.99	0.93	0.81	0.91	0.95
MR24-021-829	1.0	0.86	0.97	0.68	0.83
MR24-021-830	0.98	0.91	0.85	0.75	0.95
MR24-021-833	1.0	0.9	0.96	0.94	0.97
MR24-021-834	1.0	0.9	0.94	0.94	0.98

Model Correlation Coefficient Table

index	IGG	VCC	Viability	Glutamate	Ammonium
MR24-021-825	1.0	0.75	0.98	0.97	0.99
MR24-021-826	1.0	0.92	0.98	0.76	0.99
MR24-021-827	1.0	0.93	0.99	0.99	1.0
MR24-021-828	1.0	0.94	0.99	0.96	1.0
MR24-021-829	1.0	0.89	0.98	0.99	1.0
MR24-021-830	1.0	0.92	0.98	0.98	1.0
MR24-021-833	1.0	0.9	0.99	0.98	0.99
MR24-021-834	1.0	0.91	0.97	0.98	0.99